

Loading dataset

```
dpath = 'restaurantLab_ExtMR.xlsx'; % dataset path

T = readtable(dpath, 'ReadVariableNames',true, 'ReadRowNames',true);

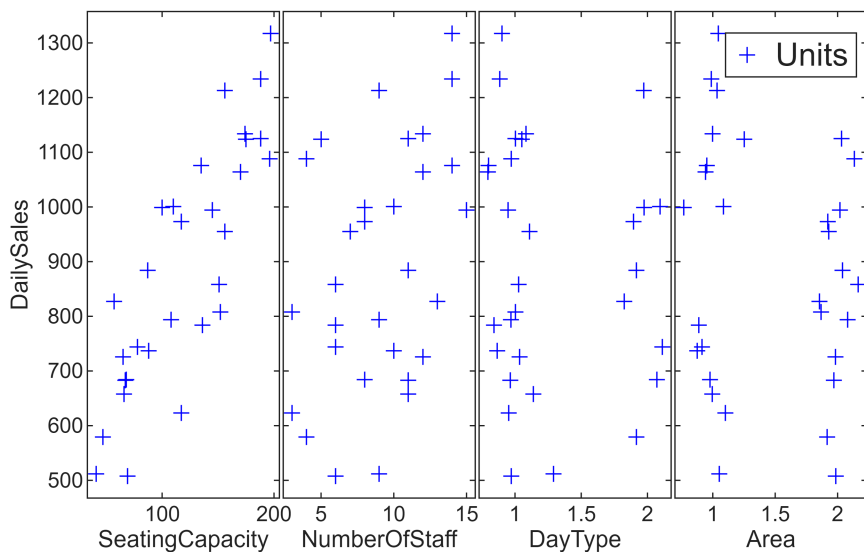
[n, p] = size(T); % n = no of rows, p = no of columns
```

Converting categorical variables.

```
T.DayType = categorical(T.DayType, 'Ordinal', true);
T.Area = categorical(T.Area, 'Ordinal', true);
```

Plotting Daily Sales

```
yXplot(T(:, "DailySales"), T(:, 2:end));
```



The yXplot shows that the **SeatingCapacity** has the strongest relationship with **DailySales**, which obviously means bigger outlets make more daily sales than small ones. Moreover, it does not illustrate a lot of Area and DayType.

Full Multiple Regression Model

```
% fitting the model
mdl = fitlm(T, "DailySales");

disp(mdl); % model summary
```

Linear regression model:
DailySales ~ 1 + SeatingCapacity + NumberOfStaff + DayType + Area

Estimated Coefficients:

	Estimate	SE	tStat	pValue
(Intercept)	118.19	43.82	2.6971	0.012342
SeatingCapacity	4.3516	0.22473	19.363	1.4485e-16
NumberOfStaff	22.671	2.9024	7.8113	3.6207e-08
DayType_weekend	178.17	23.486	7.5863	6.1006e-08
Area_suburb	-19.295	20.026	-0.96348	0.34454

Number of observations: 30, Error degrees of freedom: 25
Root Mean Squared Error: 53.8
R-squared: 0.949, Adjusted R-Squared: 0.941
F-statistic vs. constant model: 117, p-value = 8.64e-16

All other predictors statistically significant except the Area. Because, the p-value > 0.05 of Area for suburb.

Removing insignificant variables

```
% remove Area
mdl1 = removeTerms(mdl, "Area");

disp(mdl1); % new model summary
```

Linear regression model:
DailySales ~ 1 + SeatingCapacity + NumberOfStaff + DayType

Estimated Coefficients:

	Estimate	SE	tStat	pValue
(Intercept)	102.22	40.511	2.5233	0.018076
SeatingCapacity	4.3785	0.22269	19.662	3.947e-17
NumberOfStaff	23.019	2.8759	8.0042	1.7518e-08
DayType_weekend	180.13	23.365	7.7096	3.5129e-08

Number of observations: 30, Error degrees of freedom: 26
Root Mean Squared Error: 53.7
R-squared: 0.947, Adjusted R-Squared: 0.941
F-statistic vs. constant model: 156, p-value = 9.93e-17

Even after removing **Area** variables the model still explains the sales well. The model illustrates, an **Outlet** on a weekday with average **Seating Capacity** and typical **number of staff** produce sales of roughly **102** meals. However, on **+1** increase of seating capacity increases sales by **4** four additional sales. And increasing the staff by **+1** will increase sales with **+23** more.

Testing Joint Hypothesis

```
% Define the restrictions for the coefficients
C = zeros(2, 4);

C(1, 2) = 1; % beta_noOfStaff
C(2, 3) = 1; % beta_dayType

[pVal, fTest] = coefTest(mdl1, C);

% Display the p-value and F-test statistic
```

```
fprintf('p-value: %.16f \nF-test statistic: %.4f\n', [pVal, fTest]);
```

```
p-value: 0.0000000000000000  
F-test statistic: 233.2459
```

The p-value is extremely small and F-statistics is around 233 which clearly explains that they are strongly significant.

Stepwise Selection and Comparison

This method automatically select a subset of independent variables that optimally predict the dependent variable.

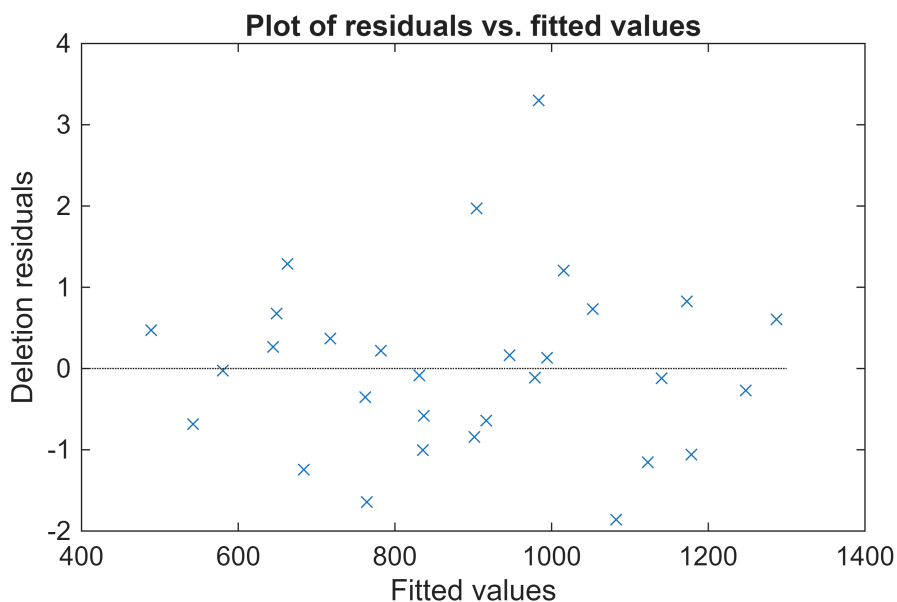
```
swMdl = stepwiselm(T, "DailySales");
```

1. Adding SeatingCapacity, FStat = 69.2876, pValue = 4.67646e-09
2. Adding NumberOfStaff, FStat = 17.851, pValue = 0.000243432
3. Adding DayType, FStat = 59.4385, pValue = 3.51295e-08

The **stepwise model** and the **reduced model** are completely identical with each other.

Plotting deletion (studentized) residuals against fitted values

```
close all; % just to prevent plotting errors (haha)  
  
plotResiduals mdl1, "fitted", "ResidualType", "studentized";  
xlabel('Fitted values');  
ylabel('Deletion residuals');
```



There is a **curvature** shape made by data points with rare outliers. Most of them are in range of **-1 to 2**, they are not outliers.

Model Prediction

```

% to predict daily sales we need,
% 1. seating capacity; 2. no of staff; 3. day type;

x0 = T(3, 2:4); % we used T, just for labels and making it a table
x0.SeatingCapacity = 160;
x0.NumberOfStaff = 8;
x0.DayType = "weekend";

% Predict daily sales using the fitted model
[predictedSales, predCI] = predict mdl1, x0, 'Alpha', 0.01, ...
    'Prediction', 'Observation');

fprintf('Predicted Daily Sales: %.2f\n', predictedSales);

```

Predicted Daily Sales: 1167.06

```

fprintf('99 per cent prediction Confidence Interval:\n [%.4f, %.4f]\n', predCI(1),
predCI(2));

```

99 per cent prediction Confidence Interval:
[1003.8532, 1330.2679]